Grade 9 Physics – Summary

The key facts, formulas and diagrams on a few pages. Use your full notes for the detailed explanations.

Formulas

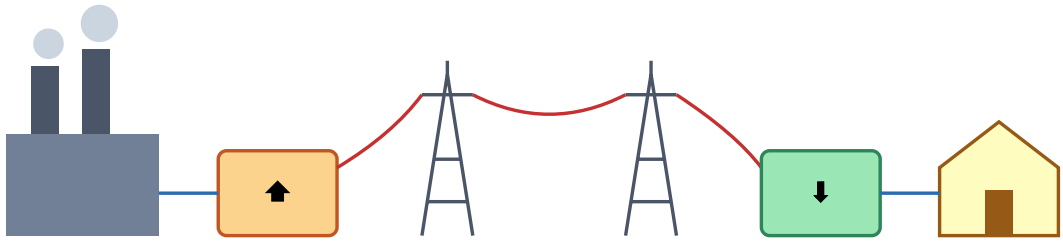
Formula	Units	Remember
Q = I × t	C = A × s	Change minutes to seconds (× 60)
V = I × R	V = A × Ω	R = V ÷ I; I = V ÷ R
W = V × Q	J = V × C	Energy a cell gives the charge
w = m × g	N = kg × N/kg	g = 10 N/kg; change grams to kg (÷ 1 000)
E = P × t; cost = kWh × price	kWh = kW × h	Change W to kW (÷ 1 000)

Every calculation: formula → substitute → answer with unit.

Electricity

Cells and generating electricity

- **Cell:** chemical energy → electrical energy. **Battery:** two or more cells; in series their voltages add up.
- **Power station:** fuel/energy source → turbine turns → **generator** (magnets + coils of wire) → electricity.
- **Coal:** chemical → heat → kinetic → electrical (most SA electricity; pollutes, non-renewable). **Nuclear** (Koeberg), **hydro**, **wind**, **solar** (light → electrical, no turbine).
- **Renewable:** sun, wind, water. **Non-renewable:** coal, oil, gas, uranium.

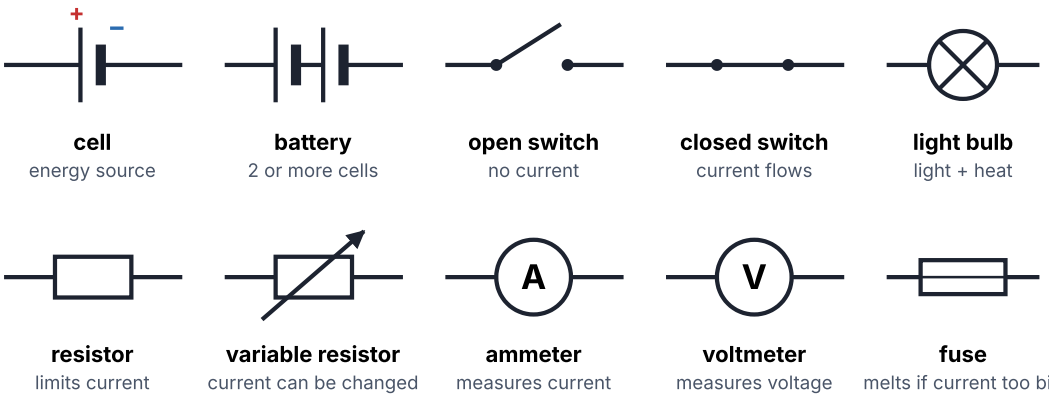


High voltage → small current → less energy lost as heat in the long wires

How electricity gets from the power station to your home

- **National grid:** power station → step-up transformer (high voltage, up to 400 000 V) → pylons → substation (step-down) → homes at 230 V. High voltage = small current = less energy lost as heat.
- **Cost:** 2 000 W heater for 3 h at R2,50/kWh → 2 kW × 3 h = 6 kWh → 6 × R2,50 = **R15**.

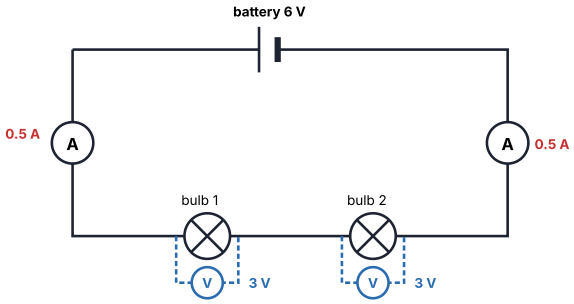
Circuit components and key terms



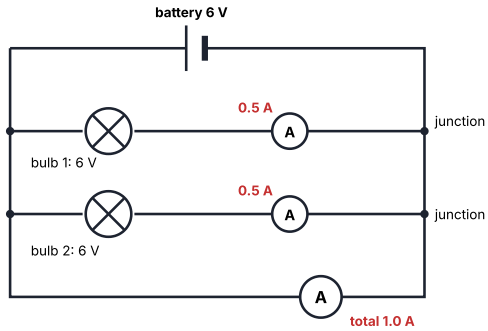
Circuit symbols you must know. Draw circuits with a ruler, straight lines and square corners.

Term	Meaning	Unit	Measured with
Charge (Q)	Carried by electrons; not used up	coulomb (C)	–
Current (I)	Rate of flow of charge (charge per second)	ampere (A)	Ammeter – in **series**
Potential difference (V)	Energy per coulomb of charge between two points	volt (V)	Voltmeter – in **parallel**
Resistance (R)	Opposition to the flow of current	ohm (Ω)	R = V ÷ I

Series and parallel circuits



A series circuit. Both ammeters read the same (0.5 A). The 6 V is shared between the bulbs: 3 V + 3 V = 6 V.



A parallel circuit with two branches. Each bulb gets the full 6 V. The current splits: 0.5 A + 0.5 A = 1.0 A from the battery.

	Series	Parallel
Current	Same everywhere	Splits: $I = I_1 + I_2$
Voltage	Shared: $V = V_1 + V_2$	Same across each branch
Resistance	Adds up: $R = R_1 + R_2$	Total gets smaller
Add a bulb	Dimmer	Same brightness
One bulb blows	All go out	Others stay on

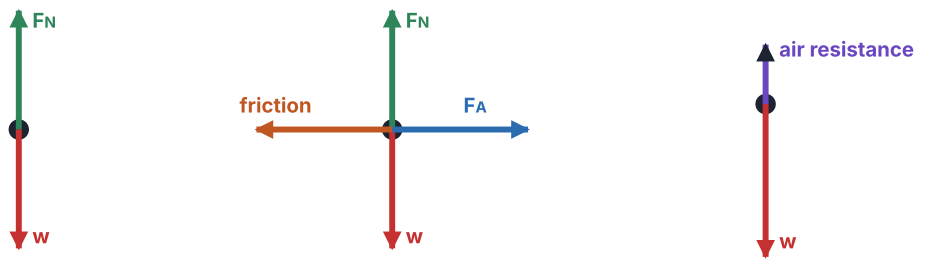
Series: 12 V with 2 Ω and 4 Ω → R = 6 Ω → I = 12 ÷ 6 = 2 A → V_1 = 4 V, V_2 = 8 V.
Parallel: 6 V with 3 Ω and 6 Ω → I_1 = 2 A, I_2 = 1 A → total 3 A.

What affects resistance

- **Longer** wire → more resistance. **Thicker** wire → less. **Hotter** → more. **Material:** copper low, nichrome high.
- **Conductors** (metals) let current flow; **insulators** (plastic, rubber, wood, glass) do not.

Forces

- **Force:** a push or pull; unit **newton (N)**; drawn as an arrow (size + direction).
- **4 effects:** change shape · start or stop motion · change speed · change direction.
- **Contact forces** (touching): applied force, friction (opposes motion), normal force (90° to surface), tension (along a rope).
- **Field forces** (at a distance): weight (gravity, straight down), electrostatic, magnetic.



Book resting on a table
forces balanced

Car at constant speed
FA = friction (equal arrows)

Ball falling, speeding up
weight bigger than air resistance

Free body diagrams for three common exam situations

Free body diagram: object as a dot, arrows from the dot, longer arrow = bigger force, label every arrow, use a ruler.

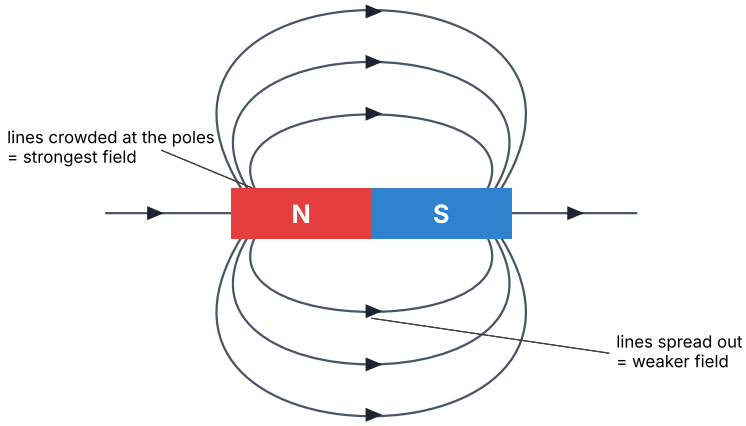
	Mass	Weight
Meaning	Amount of matter	Earth's pull on the object
Unit / instrument	kg / balance	N / spring balance
On the Moon	Same	About ⅙

w = m × g: 45 kg → 45 × 10 = **450 N**; 800 N → m = 800 ÷ 10 = **80 kg**; 250 g = 0,25 kg → **2,5 N**.

Magnetism

- Every magnet has an **N** and an **S** pole; force is strongest at the poles. **Like poles repel, unlike poles attract.**
- **Magnetic field:** region around a magnet where a magnetic force is felt. Field lines go **N → S**, never cross, are closest at the poles.

- **Ferromagnetic:** iron, nickel, cobalt (and steel). Copper, aluminium, gold, plastic, wood are not magnetic.

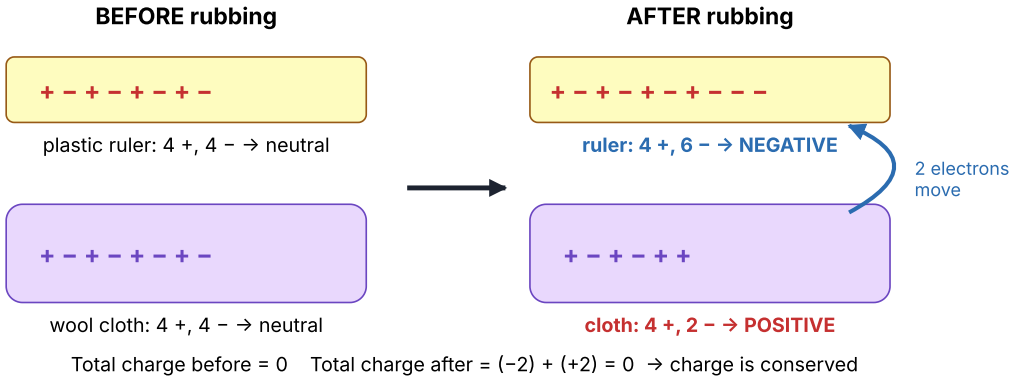


Magnetic field of a bar magnet. The arrows go from N to S outside the magnet.

- **Earth's field:** Earth acts like a giant bar magnet (molten iron in the outer core). Its magnetic **south** pole is near the geographic North Pole, so a compass's N end points north. It protects us from the solar wind and causes the aurora.

Electrostatics

- Protons (+) stay in the nucleus; **only electrons (−) move**. A **neutral** object has equal + and −.
- **Rubbing (friction)** moves electrons: object that **gains** electrons → **negative**; object that **loses** electrons → **positive**. (Ruler + wool: ruler −, wool +.)
- **Like charges repel, unlike charges attract.** A charged object also attracts neutral objects (comb picks up paper).
- **Conservation of charge:** charge cannot be created or destroyed, only transferred; the total stays the same.



Rubbing a plastic ruler with a woollen cloth. Electrons move from the cloth to the ruler.